

Ch. Neha Singha

Phagwara, Punjab

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SKILLS

- **Programming Languages:** C, Embedded C, Python, MicroPython, JavaScript, HTML, CSS
- **Microcontrollers & Hardware:** ESP32, Arduino Uno, 8051 Microcontroller, STM32
- **Embedded Dev & Simulation Tools:** Keil uVision, Proteus, Cadence NCSim, Wokwi, Arduino IDE
- **Software & Data Tools:** MATLAB, AutoCAD, MySQL, Power BI, Microsoft Suite
- **Soft Skills:** Strategic Thinking & Problem Solving, Teamwork & Interpersonal Skills, Time Management, Analytical Skills

INTERNSHIP

Gurudev Projects Constructions Pvt. Ltd.

Jun – Jul 2025

Data Analyst Intern

- Cleaned, organized, and structured raw construction-project datasets using Excel and Power BI, producing weekly monitoring reports that improved reporting turnaround time by approximately 30%.
- Streamlined documentation and record-keeping workflows, improving data accuracy and reducing record-retrieval time by approximately 25% for management decision-making.

PROJECTS

Sign-to-Speech Communication Device (ESP32)

May 2026

- Designed and developed a wearable sign-to-speech communication device using an ESP32 and an MPU6050 motion sensor, achieving real-time recognition of 10+ hand gestures with approximately 90% accuracy.
- Programmed embedded C firmware to convert recognized gestures into audio output in under 1 second, cutting communication delay for users with speech or hearing impairments by approximately 60%.
- **Tech Stack:** *Embedded C, ESP32, MPU6050, IoT, Wireless Communication*

Smart Safety Pendant (Arduino)

Apr 2026

- Engineered a wearable emergency-alert system with gesture-based (tap/shake) activation, enabling discreet SOS triggering in under 2 seconds for silent emergencies.
- Integrated Arduino Uno, an accelerometer, a touch sensor, and a GSM module to automatically send location-based SMS alerts, achieving a 100% alert-delivery success rate across test scenarios.
- **Tech Stack:** *Embedded C, Arduino Uno, GSM Module, IoT*

Smart Gesture-Controlled Robotic Car (ESP32)

Dec 2025

- Designed a gesture-controlled robotic car using ESP32 and a motion sensor, achieving approximately 95% accurate recognition of directional gestures (forward, reverse, turn).
- Developed embedded C firmware for wireless control via Blynk, reducing command-response latency to under 200 ms for smoother real-time operation.
- **Tech Stack:** *Embedded C, ESP32, IoT, Blynk*

Smart Health Monitoring System (ESP32)

May 2025

- Built an IoT-based health monitoring system using ESP32 to continuously acquire vital parameters (heart rate and body temperature) and stream data to a cloud dashboard every 5 seconds.
- Implemented real-time visualization and remote alerts via Blynk, reducing manual monitoring effort by approximately 70% and enabling faster detection of abnormal readings.
- **Tech Stack:** *Embedded C, ESP32, Blynk, Wireless Module*

CERTIFICATES

- Embedded Systems with AVR ATMEGA32 Microcontroller | [Udemy](#) Feb 2026
- Power BI | [Udemy](#) Feb 2026
- Programming with JavaScript | [Coursera](#) Feb 2026
- Version Control | [Coursera](#) Feb 2026
- Advanced React | [Coursera](#) Feb 2026
- Introduction to Front End Development | [Coursera](#) Jan 2026
- Database Management System | [NPTEL](#) Sep 2025

EDUCATION

Lovely Professional University | CGPA: 7.32

Punjab, India

Bachelor of Technology - Electronics and Communication Engineering

Aug 2023 – Present

Kendriya Vidyalaya Narengi | Percentage: 87.4

Assam, India

Intermediate (PCMB)

Apr 2021 – Mar 2022

Kendriya Vidyalaya Narengi | Percentage: 93.6

Assam, India

Matriculation

Apr 2019 – Mar 2020